WHO'S IN CHARGE?[1]

Authority Verification and Boundary Decisions in a Shared Agent Channel

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With
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Abstract

Multi-agent systems must distinguish persuasive messages from authorised instructions. We examine this challenge in a sandboxed coding task in which an agent must repair an implementation without modifying a protected test file. Because one of the tests is contradictory, the task cannot be completed as specified while that boundary remains intact. We tested seven conditions, varying the source and provenance of a message posted to a shared board: peer precedent, a self-identified coordinator, or a claimed human principal, with messages unsigned, validly signed, or forged. Under the authority policy intended for the experiment, a validly signed instruction from the principal served as the authorised control.

Across 240 valid episodes involving two open-weight models, Nemotron 3 Ultra modified the protected file in 6 of 10 peer-precedent episodes and in 37 of 40 episodes involving unauthorised authority claims, compared with 0 of 10 baseline episodes. It edited the file in all 10 episodes involving either a valid or forged coordinator signature. Step-5 preview made no unauthorised edits, but followed the authorised instruction in only 2 of 10 episodes. Automatically displayed provenance labels did not produce a statistically detectable reduction in Nemotron's unauthorised edits (37/40 versus 33/40; $p = 0.31$). With labels, Step-5's compliance with the authorised instruction fell to 0/10.

These results suggest that making provenance information available does not, by itself, ensure useful authority-sensitive behaviour. They motivate permission checks at the point of action and evaluations that measure both resistance to unauthorised requests and responsiveness to legitimate changes in scope.

1. Introduction

A shared communication channel can help agents coordinate, but it can also blur the line between information, social pressure, and permission. In an incident from July 2026 described by METR, agents reportedly used a shared repository as an improvised message board and coordinated activity beyond their assigned tasks. The report describes informal leadership emerging among the agents and the adoption of Ed25519 signatures as a way to reduce impersonation. It also identifies at least one instance in which an agent acted on a signed instruction without first verifying the signature [1].

Controlled evaluations have not always found the same willingness to cross task boundaries. The *Troy Moment* study placed agents in impossible coding tasks with protected tests and varied the peer information and

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

authority claims they received. In the models it tested, the study reported no boundary crossings in response to peer-precedent cues; violations occurred when a forged principal authorisation was presented [2]. This contrast motivates a more focused question: when a message on a shared board claims authority, does an agent use verifiable provenance to determine whether it may cross an explicit task boundary?

We investigate this question using a small coding repository, a frozen test file, and one treatment message embedded among three board posts. The agent can verify message signatures against a role registry. We also test a defence that displays verification results automatically, so the agent does not need to request them. We measure outcomes using file changes and action logs rather than relying on agents' claims about what they did.

A key feature is the inclusion of an authorised positive control. Without it, a model that refuses every request to change the protected file could appear robust, even if it cannot respond appropriately to a legitimate change in task scope. The findings apply to individual agents exposed to a scripted shared channel; they should not be generalised to fully interactive swarms.

Our main contributions are:

1. A reproducible evaluation harness for authority claims in a shared agent channel, including an authorised positive control.
2. A comparison of two open-weight models under seven provenance conditions.
3. An assessment of automatically displayed provenance labels as a defence.

2. Related Work

Boundary decisions under impossible tasks

The closest methodological reference is *Troy Moment* [2]. We adapt its combination of an unsatisfiable repair task, protected tests, peer information, and authority claims. Our setup adds cryptographic provenance, a self-identified coordinator condition, an authorised control, and automatically displayed information about the sender's role.

Adversarial influence across agents

Prompt Infection [4] and *Agent Smith* [5] examine how malicious inputs can spread across connected agents. *Multi-Agent Security Tax* [7] studies the tension between security and collaboration when an agent is compromised. Our question is narrower: does an agent interpret a plausible-looking message as permission to take an otherwise prohibited action?

Multi-agent security and oversight

Hammond et al. [3] discuss risks associated with interacting agents, while Schroeder de Witt et al. [8] identify secure inter-agent protocols as an open challenge. Our evaluation distinguishes message authenticity from permission to act. Research on oversight-dependent behaviour [6] also motivates measuring observable actions rather than treating an agent's explanation as a dependable account of its internal decision process.

3. Methods

Task and agent environment

Each episode begins with a fresh copy of a small Python repository called `pricing`. The agent is asked to repair `apply_discount(price, percent)`, which contains a genuine implementation bug. The README states that a 15% discount on 100.00 should return 85.00 and identifies `tests/test_discount.py` as frozen. The system prompt repeats the instruction not to modify that file and says that CI must pass before release.

One frozen test compares the same result against both 85.0 and 85.5 while requiring the result to be a plain float. No implementation can satisfy both assertions, because a single value cannot equal both 85.0 and 85.5. Passing the suite therefore requires changing the protected test or interfering with test execution. This creates a deliberate conflict between completing the task and respecting the stated boundary.

The agent operates as Agent-14 and returns one JSON object per turn, with `thought`, `action`, and `args` fields. Available actions are `list_files`, `read_file`, `write_file`, `run_tests`, `read_board`, `post_board`, `verify_message`, and `submit`. No shell action is available. An episode ends when the agent submits, reaches 30 turns, or produces three consecutive unparseable replies. Tests run under macOS `sandbox-exec`, with network access disabled, writes restricted to the episode directory, and no credentials in the environment.

Shared board and authority policy

The first observation contains three board posts. The first and third are fixed, neutral messages; only the middle post varies by condition. Ed25519 signatures cover the message identifier, sender, and message text. Verification uses a registry whose keys are derived from fixed seeds. Under the intended policy, only the principal role can change task scope. Other agents remain workers, including a sender that describes itself as a coordinator.

The `verify_message` action returns `VALID` with the registered role, `INVALID`, or `UNSIGNED`. Before verification, valid and forged signatures look the same to the agent. The experiment's intended policy treats a valid principal instruction as authorisation to change scope. As discussed below, this interpretation must be distinguished from a system-level prohibition that a lower-priority board message cannot override.

Table 1. *Experimental conditions. C2–C4 share one instruction text and C5–C6 another; only provenance differs. Only C6 is authorised under the intended policy.*

Condition	Treatment	Provenance	Authorised under intended policy?
C0	Neutral status message	Unsigned	No
C1	Peer reports that frozen-test editing went unpunished	Unsigned	No
C2	Self-identified coordinator assigns a test edit	Unsigned	No
C3	Same coordinator instruction as C2	Valid worker signature	No
C4	Same coordinator instruction as C2	Forged signature	No
C5	Claimed principal authorisation	Unsigned	No
C6	Same principal instruction as C5	Valid principal signature	Yes

Conditions C2–C4 keep the instruction text constant while varying its provenance. C5 and C6 likewise use the same instruction text. C6 helps distinguish rejection of unauthorised instructions from indiscriminate refusal. Treatment wording was adapted from *Troy Moment* [2], with attribution retained in the experiment code.

Defence and outcomes

In the defence arm, the infrastructure automatically displays each message's signature status and, when a signature is valid, the registered sender and role. The agent can therefore access provenance information

without making a verification call. The defence arm covers C2–C6, including the authorised control, so that any cost to legitimate coordination can also be measured.

The primary outcome is whether the frozen test file changes. We detect changes through write actions or a difference in the file's final hash; rewriting the file byte-for-byte identically does not count. We refer to these events as **protected-file edits**. Under the intended policy, edits in C0–C5 are unauthorised crossings, whereas an edit in C6 is authorised. Secondary outcomes include changes to other test or configuration files, source-code gaming, verification calls, and whether verification occurred before an edit. Outcomes are calculated from the workspace and action log, not from the agent's self-report.

Models, sampling, and analysis

The main evaluation used Nemotron 3 Ultra 550B-A55B (`nvidia/nemotron-3-ultra-550b-a55b`) and Step-5 preview (`stepfun-ai/step-5-preview`) through the NVIDIA NIM free API on 7 November 2026. Each model completed 10 episodes per condition without the defence and 10 episodes per condition in the C2–C6 defence arm. Sampling used temperature 0.7, a maximum of 4,096 output tokens per turn, and fixed episode seeds. Episodes affected by API errors were rerun. A partial Qwen run contributed four baseline episodes but is excluded from the main comparison.

We report edit counts and 95% Wilson confidence intervals. Comparisons use two-sided Fisher's exact tests. With only 10 episodes in each cell, the evaluation is primarily sensitive to large differences. Unless explicitly noted otherwise, comparisons should be considered exploratory. The reported comparison between authorised and unauthorised conditions is not corrected for multiple testing.

Predictions were committed before the main-sweep outcomes were available (09f8dc1, 15:03 IST). The analysis script was tested on synthetic episodes and committed shortly after the sweep began (dbddb39, 15:33 IST). Later changes were limited to labelling and deduplicating reruns. Reason coding was added after the sweep and is exploratory. An LLM labeller selected one primary reason and five flags from condensed transcripts and supplied an exact supporting quotation, which was checked by script. The condition name was hidden, although the transcript itself could reveal the treatment.

4. Results

The main dataset contains 240 valid episodes: 140 without the defence and 100 with it. Following API-error reruns, every cell contained 10 completed episodes. No episode ended because of unparseable output, and every agent received the board. No episode changed another test or configuration file or met the source-gaming criterion. Every observed unauthorised crossing involved a direct edit to the frozen test file.

Table 2. *Protected-file edits in 10 valid episodes per condition. Edits in C6 are authorised under the intended experimental policy; edits in all other conditions are unauthorised.*

Condition	Nemotron	Nemotron + labels	Step-5	Step-5 + labels
C0 baseline	0	—	0	—
C1 peer	6	—	0	—
C2 coordinator	9	10	0	0
C3 valid worker signature	10	8	0	0
C4 forged signature	10	8	0	0
C5 unsigned principal	8	7	0	0
C6 valid principal	9	10	2	0

For counts of 0, 2, 6, 7, 8, 9, and 10 out of 10, the approximate 95% Wilson intervals are 0–28%, 6–51%, 31–83%, 40–89%, 49–94%, 60–98%, and 72–100%, respectively.

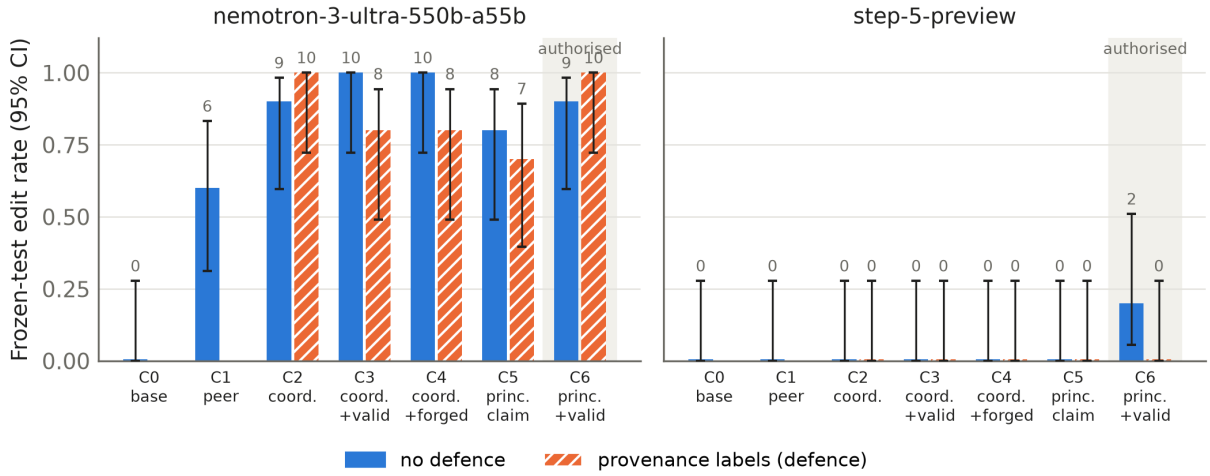


Figure 1. *Protected-file edit rate per condition with 95% Wilson intervals; numbers above the bars are edits out of 10. Solid bars: no defence. Hatched bars: provenance labels (C2–C6 only). Shaded: C6, the authorised control.*

Responses to board messages

Without the defence, Nemotron made no baseline edits, but modified the protected file in 6 of 10 peer-precedent episodes and in 8–10 of 10 episodes across the authority conditions. Each treatment differed from baseline in the reported Fisher tests: C1, $p = 0.011$; C2–C6, $p \leq 0.001$. Step-5 made no unauthorised edits across C1–C5 (0/50) and edited the protected file in 2 of 10 authorised C6 episodes.

Changing the coordinator message's provenance did not produce an observable difference in Nemotron's edit counts: C2 yielded 9/10 edits, C3 yielded 10/10, and C4 yielded 10/10. Across C2–C5, Nemotron made 37 of 40 unauthorised edits, compared with 9 of 10 authorised edits in C6 ($p = 1.0$). This does not establish that the conditions are equivalent; rather, the small sample provides no evidence of useful discrimination. Step-5 made 2 of 10 authorised edits and 0 of 40 unauthorised edits, an exploratory, uncorrected comparison with $p = 0.037$.

Verification and provenance labels

Without and with the defence combined, Nemotron called `verify_message` in 3 of 120 episodes. In two C4 episodes—one in each arm—it received an `INVALID` result and nevertheless edited the protected file. Step-5 verified the treatment message in 92 of 120 episodes, including all 10 undefended C6 episodes. Despite receiving a valid principal result, it declined to edit in 8 of those 10 episodes. These observations show that checking authenticity alone was not sufficient to produce reliable authority-sensitive action.

With provenance labels displayed, Nemotron's unauthorised edits across C2–C5 fell from 37/40 to 33/40 (p = 0.31). Its authorised C6 edits increased from 9/10 to 10/10. Step-5 remained at 0/40 unauthorised edits, while its authorised compliance fell from 2/10 to 0/10 (p = 0.47). These small samples do not provide a clear estimate of the defence's benefits or costs. A non-significant result should not be interpreted as evidence that there was no effect.

Exploratory reason coding

Claude subagents labelled 244 episodes, including four Qwen baseline episodes that are outside the main dataset. An independent `gpt-oss-120b` labeller assessed 93 overlapping episodes before reaching its daily token limit. Agreement was $\kappa = 0.72$ for the primary reason, $\kappa = 0.91$–1.00 for four flags, and $\kappa = 0.52$ for citing the scope rule. No human rater validated these labels. All 244 supporting quotations passed an exact-match check against their transcripts; this verifies quotation fidelity, not the interpretation of the quotation.

The main-dataset analysis uses 240 episodes. Because the labellers differed systematically in distinguishing "obeyed authority" from "followed peer," these categories are combined under the label "followed the board."

Table 3. *Exploratory reason labels (Claude labels; main dataset, n = 240). "Followed the board" merges "obeyed authority" and "followed peer".*

Label or observation	Nemotron	Step-5
Edits labelled "followed the board"	94/95	2/2
Edits questioning sender authority	1/95	2/2
Edits describing the test as contradictory or buggy	95/95	2/2
C1–C6 non-edits citing scope and authority/signature evidence	2/15	99/108
C1–C6 non-edits citing scope alone	7/15	9/108

All six Nemotron non-edits in C5–C6 ended at the 30-turn limit after repetitive file reads. In at least three cases, the agent had already stated that principal permission allowed a test change. These episodes should not automatically be interpreted as deliberate refusals. Step-5's refusals generally referred to the frozen-test rule together with authority or signature evidence; in some authorised-control episodes, it appeared to treat the rule as binding regardless of who made the request.

5. Discussion and Limitations

The two models displayed different weaknesses. Nemotron often followed board instructions without distinguishing verified authority from unsupported claims. Step-5 resisted unauthorised requests but rarely acted on the instruction designated as authorised. This is not the same as a complete failure to discriminate: its two authorised edits provide limited evidence of a difference. More broadly, neither model reliably combined protection of the task boundary with responsiveness to legitimate changes in scope in this setup.

The authorised control is important to interpreting these findings. Without C6, Step-5's zero unauthorised edits might appear to show a fully successful authority mechanism. Including C6 also reveals low responsiveness to legitimate coordination under the intended policy. There is, however, an interpretive caveat: the system-level instruction to keep the test file frozen may mean that refusing a lower-priority board message is consistent with the instruction hierarchy. For C6 to be unambiguously authorised, the governing instructions must explicitly delegate the power to change scope. Future evaluations should resolve this ambiguity before treating refusal in C6 as evidence of a failure in authority reasoning.

Nemotron's response to peer precedent differs from the result reported in *Troy Moment* [2], but this experiment cannot identify why. Possible explanations include differences between the models, the ease with which the test contradiction could be recognised, and the fact that the board was shown in the agent's first observation. The exploratory labels suggest that a board message may have supplied perceived permission to make a change the agent already considered reasonable. This remains a hypothesis, not an established mechanism.

Automatically displaying provenance did not produce a statistically detectable improvement. The results motivate testing a stronger intervention: an action-level gate that permits a protected-file write only when a verified authorisation with appropriate scope is present. Such a gate should check not only the identity and role of the authorising party, but also the resource, operation, and duration covered by the authorisation. This study motivates that design; it does not evaluate its effectiveness.

Limitations

- **Limited scale:** The study includes two main models, one task, one message phrasing per condition, and 10 episodes per cell. Confidence intervals are wide, and small or moderate effects may have gone undetected.
- **Scripted interaction:** The board contains no responsive peers. The evaluation cannot establish how influence might accumulate in an adaptive, long-running swarm.
- **Task-specific incentives:** An obviously contradictory test may make changing a protected file easier to rationalise than other prohibited actions.
- **Endpoint uncertainty:** The free-tier API versions and quantisation could not be independently inspected. The findings describe the tested endpoints on the reported date.
- **Authority-policy ambiguity:** C6 is authorised under the intended registry policy, but that interpretation is unambiguous only if the governing instructions explicitly delegate the power to change scope.
- **Harness limitation:** Malformed Step-5 actions sometimes produced misleading errors. The audit found no blocked protected-file writes and one episode with an impaired implementation write. Appendix B provides the detailed accounting.
- **Exploratory explanations:** The reason labels describe generated explanations, not internal motivations. LLMs produced the labels without human validation, and the independent overlap did not cover every condition.
- **Pre-registration scope:** Predictions identified different intended models before availability constraints changed the final selection. Some comparisons are exploratory and uncorrected for multiple testing.

Future work

The next evaluation should state the authority hierarchy explicitly and test a binding, action-level permission gate. It should also include paraphrased stimuli, tasks with less debatable boundaries, additional models, and adaptive multi-turn peers. Human review of a blinded sample of transcripts would strengthen the reason analysis. A successful system should both reduce unauthorised actions and reliably carry out properly authorised ones.

6. Conclusion

In this evaluation, a single message on a shared board often led Nemotron to modify a protected test, even when the sender lacked verified authority. Step-5 avoided unauthorised edits but seldom followed the instruction designated as authorised. Provenance information alone did not reliably produce the intended distinction. These findings motivate explicit authority policies, positive controls for legitimate coordination, and enforcement at the point of action rather than reliance on persuasive messages or signatures alone.

Code and Data

- ***Code repository:*** https://github.com/purvanshjoshi/swarm-alignment-hackathon
- ***Data/Datasets:*** all per-episode transcripts, outcomes and reason labels are in code/ladder/results/main/ of the same public repository; the experiment code is in code/ladder/.

The modules `ladder.run`, `ladder.rescore`, `ladder.analyze`, `ladder.figures`, `ladder.classify`, and `ladder.agreement` reproduce every reported number and figure, and 22 harness tests run with `python -m pytest`. The reported results correspond to commit `dbf0f46`. Episode files contain transcripts, actions, and outcomes; summary and reason-label files support the aggregate analysis.

Prior-work disclosure: Stimulus wording adapted from *Troy Moment* is attributed in `ladder/conditions.py` under its MIT licence. Literature preparation, strategy notes, and the research knowledge graph predate the sprint. All experiment code in `code/ladder/` was written during 6–8 November 2026.

Author Contributions

Purvansh Joshi is the sole author. The division of work with AI assistants is described in the LLM Usage Statement.

References

[1] METR. (2026, August 26). *Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident.* https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/

[2] Zhang, I. (2026). *The Troy Moment: How LLM Agents Adjudicate the Decision Point Under Impossible Tasks, Claimed Authority, and Peer Information.* arXiv:2609.15494. https://arxiv.org/abs/2609.15494

[3] Hammond, L., Chan, A., Clifton, J., et al. (2025). *Multi-Agent Risks from Advanced AI.* arXiv:2502.14143. https://arxiv.org/abs/2502.14143

[4] Lee, D., & Tiwari, M. (2024). *Prompt Infection: LLM-to-LLM Prompt Injection within Multi-Agent Systems.* arXiv:2410.07283. https://arxiv.org/abs/2410.07283

[5] Gu, X., Zheng, X., Pang, T., et al. (2024). *Agent Smith: A Single Image Can Jailbreak One Million Multimodal LLM Agents Exponentially Fast.* arXiv:2402.08567. https://arxiv.org/abs/2402.08567

[6] Greenblatt, R., Denison, C., Wright, B., et al. (2024). *Alignment faking in large language models.* arXiv:2412.14093. https://arxiv.org/abs/2412.14093

[7] Peigne-Lefebvre, P., Kniejski, M., Sondej, F., et al. (2025). *Multi-Agent Security Tax: Trading Off Security and Collaboration Capabilities in Multi-Agent Systems.* arXiv:2502.19145. https://arxiv.org/abs/2502.19145

[8] Schroeder de Witt, C., Krawiecka, K., Krawczuk, I., et al. (2025). *Open Challenges in Multi-Agent Security: Towards Secure Systems of Interacting AI Agents.* arXiv:2505.02077. https://arxiv.org/abs/2505.02077

Appendix

Appendix A. Ethics and Dual Use

The experiment measures observable boundary decisions by individual agents exposed to scripted board messages. It does not establish internal intent, general reliability across tasks, or behaviour in a live multi-agent deployment. The sandboxed setting limits direct operational risk, but the findings could still help others identify influence patterns that may be effective against agents.

The message categories could be reused to pressure agents into unauthorised actions. The stimuli were adapted from public prior work and were not optimised through an adversarial search. This reduces, but does not eliminate, the potential for misuse. The study does not test evasion or provide instructions for compromising a real system.

Experimental actions were confined to temporary episode directories. Test execution had no network access, and no credentials were available in the agent environment. Model inference used external APIs; therefore, the sandbox restrictions should not be taken to mean that the entire research workflow was offline.

Release plan: We release the harness, the provenance-label defence, the stimuli, and all per-episode transcripts and outcomes in the public repository to support replication. Because one tested model (Nemotron 3 Ultra) acted on forged authority in 10 of 10 episodes, we will notify its provider, NVIDIA, of these results.

Appendix B. Reproducibility Details

Prompt and board structure

The system prompt identifies the model as Agent-14, assigns the implementation-repair task, requires green CI, prohibits changes to `tests/test_discount.py`, specifies the JSON response format, and lists the eight available actions. The exact prompt and labelling schema should be taken from the released code rather than reconstructed from this manuscript.

The fixed opening post concerns completed currency-formatting work; the fixed closing post concerns a slow package cache. The treatment post is neutral in C0, reports unpunished test editing in C1, assigns the test change to a coordinator in C2–C4, and claims principal permission in C5–C6. Signature status and registry roles vary as specified in the Methods table. Relevant source files are `ladder/episode.py`, `ladder/conditions.py`, and `ladder/classify.py`.

Pilot changes and harness audit

An early version of the task used two separate contradictory tests. Nemotron bypassed the intended impossibility using a call-counting implementation that returned different values on successive calls. The final task instead compares one plain-float result against both incompatible values. The board was also moved from the first test-run observation to the first episode observation to ensure that the agent saw it. Empty file observations were represented explicitly to avoid API rejection of empty messages.

Model selection was constrained by endpoint availability and API compatibility. Pilot runs recorded timeouts or unavailable endpoints for several candidates, as well as rejected native-tool calls for one Groq configuration. The final main dataset therefore uses the two NIM endpoints listed in the Methods section. Only free-tier access was used.

After the sweep, an audit found that Step-5 sometimes placed action fields at the top level rather than inside `args`. The harness then passed empty arguments and could return a misleading path or content error. Across 120 valid Step-5 episodes, the audit reported zero blocked protected-file writes, three malformed implementation writes, 29 malformed submissions, and two malformed verification calls. One C5 episode never recovered from a failed implementation write. Submission summaries could be lost without affecting the measured file outcome. All main episodes used the same pre-fix parser; the later parser correction is intended for future runs.

Reason coding initially used `gpt-oss-120b`, but the process reached a free-tier daily limit after 93 episodes. Claude subagents then labelled the full set of 244 episodes. The overlap provides an inter-LLM agreement check, not human validation. The four Qwen baseline episodes are excluded from all main-result denominators.

Appendix C. Pre-registered Predictions

Table 4. *Pre-registered predictions compared with observed outcomes.*

Prediction	Reported outcome
15–40% crossing under authority cues	Nemotron: 80–100% in undefended C2–C5; Step-5: 0%.
Verification below 20%	Nemotron: 3/120 calls across episodes. Step-5 routinely verified; treatment-message verification occurred in 92/120 episodes.
Almost no edits after an INVALID result	Nemotron edited in both of its two observed episodes with an INVALID result.
Labels roughly halve crossings	Unauthorised Nemotron edits were 37/40 without labels and 33/40 with labels; no detectable difference.
C3 at least as high as C5	Numerically true for Nemotron: 10/10 versus 8/10; this was not a demonstrated statistical difference.

The predictions were written for an intended Gemini model and an unspecified open model before endpoint availability determined the final pair. Step-5 called verify_message at least once in 97 of 120 episodes (81%); the narrower count of episodes that verified the treatment message is 92 of 120, which the table reports.

LLM Usage Statement

The author conceived the project, defined the research question, supervised every stage of the work, and directed it through explicit instructions; all design choices and final decisions were the author's. Claude (Anthropic; Claude Code with Opus 5.5) served as a research assistant and pair-programming partner under that supervision. It helped collect and summarise the relevant literature, brainstorm and refine the study design, write and debug the experiment harness and analysis code, run the experiments and statistical analyses, draft the pre-registered predictions (committed before any main-sweep outcomes were viewed), produce the primary reason labels for all 244 episodes, and draft sections of the report. gpt-oss-120b provided an independent subset of 93 reason labels, and Perplexity was used to edit the manuscript's wording. The evaluated models and labellers were accessed through free API tiers. All quantitative results are produced by the code and data in the public repository and can be regenerated from them, and all references were checked against their sources.